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Incident Information

Field	Value
Incident ID	SOC-2025-01-22-001
Incident Type	Malicious File Download / C2 Beaconing
Severity	High
Detection Source	Manual PCAP Review (Traffic Analysis Exercise)
Analyst	Lakshmeesh R
Date Investigated	22-Jan-2025
Evidence Source	2025-01-22-traffic-analysis-exercise.pcap
Tool Used	Wireshark

Executive Summary

Manual review of packet capture traffic identified a Windows workstation retrieving suspicious files — including a PowerShell script — from an external host over plain HTTP, followed by a sustained pattern of periodic outbound polling and additional file retrieval disguised as browser-extension content.

The host first downloaded a small file and a .ps1 PowerShell script from **5.252.153.241**, then began issuing repeated GET requests to a numeric endpoint on the same host every ~5 seconds, consistently receiving HTTP 404 responses — a pattern consistent with C2 polling/beaconing for tasking. Separately, the host retrieved two additional files from **199.232.214.172** via a /filestreamingservice/ path, both served with a Content-Type: application/x-chrome-extension

header despite being fetched outside any browser extension update flow — a technique consistent with masquerading to blend malicious payload delivery in with legitimate browser traffic.

The affected workstation, its MAC address, hostname, authenticated domain user, and full identity were positively identified by correlating LLMNR, Kerberos, and SAMR traffic against the host generating this activity. The beaconing pattern persisted for the duration of the capture (over 40 minutes), indicating an active, ongoing implant rather than a one-time download.

Suspicious Activity Summary

Item	Value
Primary suspicious host	5.252.153.241 (script hosting + beacon target)
Secondary payload host	199.232.214.172 (filestreamingservice)
Destination Port	TCP/80 (HTTP)
First suspicious request	2025-01-23 01:15:56 (as captured) — GET /api/file/get-file/264872
PowerShell script retrieved	2025-01-23 01:15:58 (as captured) — /api/file/get-file/29842.ps1
Beaconing observed	Repeated GET /1517096937 every ~5 seconds, HTTP 404 responses, continuing through the full capture window

Environment

Item	Value
AD Domain / Realm	BLUEMOONTUESDAY
Domain Controller	10.1.17.2
Internal Network	10.1.17.0/24 (inferred from host address)
Gateway	Not confirmed from available evidence

Investigation Methodology

The packet capture was examined in Wireshark. The following protocols were analyzed to trace the activity from initial file download back to host and user identity:

- HTTP — malicious file download and C2 beaconing traffic
- Ethernet / IPv4 headers — IP and MAC correlation (read directly from the HTTP evidence frame)
- LLMNR — hostname discovery
- Kerberos — authenticated user account
- SAMR — full user identity

The investigation began with anomalous HTTP requests (a .ps1 download followed by repeated 404 polling to an unrelated numeric path) and worked outward to confirm host identity, user identity, and the full scope of the download/beacon activity.

Investigation Findings

1. Infected Windows Client

Field	Value
IP Address	10.1.17.215

Evidence: Filtering on http surfaced a GET /api/file/get-file/264872 HTTP/1.1 request from **10.1.17.215** to **5.252.153.241**, followed shortly after by a request for a .ps1 PowerShell script from the same external host. The IP header of this frame confirms **10.1.17.215** as the source of the file-retrieval traffic. (See Figure 1.)

2. MAC Address

Field	Value
MAC Address	00:d0:b7:26:4a:74

Evidence: The same HTTP frame used to identify the source IP also contains the Ethernet II layer, which shows Src: Intel_26:4a:74 (00:d0:b7:26:4a:74) — confirming the hardware address of the host issuing the suspicious file requests. (See Figure 1.)

3. Hostname

Field	Value
Hostname	DESKTOP-L8C5GSJ

Evidence: Filtering on llmnr surfaced repeated Link-Local Multicast Name Resolution queries originating from **10.1.17.215**, sourced from MAC **00:d0:b7:26:4a:74**. The query name resolves to **DESKTOP-L8C5GSJ**, confirming the host's Windows hostname. The same hostname is repeated in multiple query bursts across the capture (at 01:14, 01:24, and 01:25, as captured), consistent with normal periodic name broadcast behavior from the same machine. (See Figure 2.)

4. User Account

Field	Value
Username	shutchenson

Evidence: Filtering on kerberos.CNameString isolated Kerberos AS-REQ/AS-REP/TGS-REP traffic between the Domain Controller (10.1.17.2) and the workstation (10.1.17.215) within the **BLUEMOONTUESDAY** realm. The CNameString field identifies the authenticated client principal as **shutchenson**. Ticket activity for this account recurs throughout the capture (01:15, 01:20, 01:23, 01:25, and later), indicating an active, ongoing authenticated session on the host during the observed suspicious activity. (See Figure 3.)

5. User Full Name

Field	Value
Full Name	Steve Hutchenson

Evidence: SAMR (Security Account Manager Remote) traffic between 10.1.17.2 and 10.1.17.215 was reviewed. The QueryUserInfo response (Info21) contains the account's Full Name field, which resolves to **Steve Hutchenson**, tying the Kerberos-authenticated account shutchenson to a real user identity. (See Figure 4.)

6. Malicious File Download Activity

The host issued two sequential GET requests to **5.252.153.241**:

- GET /api/file/get-file/264872 → HTTP/1.1 200 OK (819 bytes)
- GET /api/file/get-file/29842.ps1 → HTTP/1.1 200 OK (555 bytes)

The second request explicitly names a .ps1 (PowerShell script) file, retrieved over plain HTTP from an API-style path with no authentication visible in the capture. This is consistent with a staged downloader delivering a PowerShell payload for local execution. (*See Figure 1 for the request that initiated this activity.*)

7. C2 Beaconing and Secondary Payload Retrieval

Immediately following the file downloads, the host began issuing repeated GET /1517096937 HTTP/1.1 requests to **5.252.153.241**, each answered with HTTP/1.1 404 Not Found. This request repeats at roughly 5-second intervals and continues for the remainder of the capture (observed as late as frame 7888 in the reviewed window, with the underlying session persisting well beyond that), consistent with a beacon polling for tasking rather than a one-off request.

Separately, the host issued HEAD/GET request pairs to **199.232.214.172** for two GUID-named resources under /filestreamingservice/files/:

- 2ed1297e-f6c9-4355-aec4-433ea371b116 → 1,171 bytes, Content-Type: application/x-chrome-extension
- 2a0d597c-a09c-4400-be86-87596dd2e696 → 1,391 bytes, Content-Type: application/x-chrome-extension

Serving arbitrary downloaded content with a browser-extension MIME type — outside of an actual browser extension update flow — is consistent with masquerading, intended to blend malicious payload delivery in with legitimate-looking web traffic.

For context, a benign GET /connecttest.txt to **23.220.102.9** was also observed early in the capture; this is a standard Windows network-connectivity check and is not part of the malicious activity.

Indicators of Compromise (IOCs)

IOC Type	Value
Internal Host	10.1.17.215
MAC Address	00:d0:b7:26:4a:74
Hostname	DESKTOP-L8C5GSJ
Username	shutchenson
Full Name	Steve Hutchenson
Primary malicious host (script + beacon)	5.252.153.241
Secondary payload host	199.232.214.172
Malicious URI (initial file)	/api/file/get-file/264872
Malicious URI (PS script)	/api/file/get-file/29842.ps1
Beacon URI pattern	/1517096937 (repeated GET, HTTP 404, ~5s interval)
Secondary payload	/filestreamingservice/files/2ed1297e-f6c9-4355-aec4-433ea371b116,

URIs	/filestreamingservice/files/2a0d597c-a09c-4400-be86-87596dd2e696
Masquerade indicator	Downloaded content served as Content-Type: application/x-chrome-extension
Destination Port	TCP/80

Timeline

(All times as captured in the local PCAP; no UTC offset is confirmed for this capture.)

Time (captured)	Event
01:14:56	First LLMNR queries observed from 10.1.17.215 (DESKTOP-L8C5GSJ)
01:15:01	Benign GET /connecttest.txt — routine Windows connectivity check
01:15:10	Kerberos AS-REQ/AS-REP for shutchenson — authenticated session begins
01:15:56	GET /api/file/get-file/264872 to 5.252.153.241 — first suspicious download
01:15:58	GET /api/file/get-file/29842.ps1 — PowerShell script retrieved
01:15:58 – onward	Beaconing begins: repeated GET /1517096937, HTTP 404, ~5s interval
01:16:23	First secondary payload retrieved from 199.232.214.172 (x-chrome-extension)
01:16:27	Second secondary payload retrieved from 199.232.214.172
01:20 – 01:26 (recurring)	Continued Kerberos ticket activity and beacon polling on the same host
01:58:00	Latest Kerberos activity observed for shutchenson in the reviewed capture

Evidence Summary

Evidence	Protocol / View	Figure
IP and MAC identification (via malicious download frame)	HTTP / Ethernet-IP headers	Figure 1
Malicious file / PowerShell download	HTTP	Figure 1
Hostname discovery	LLMNR	Figure 2
User authentication	Kerberos	Figure 3
Full user identity	SAMR	Figure 4

MITRE ATT&CK Mapping

Technique	ID	Notes
Ingress Tool Transfer	T1105	.ps1 script and additional payloads pulled from external hosts over HTTP
Command and Scripting Interpreter:	T1059.001	A PowerShell (.ps1) file was explicitly retrieved for likely local

PowerShell		execution
Application Layer Protocol: Web Protocols	T1071.001	C2 beaconing conducted over plain HTTP with a fixed polling interval
Masquerading	T1036	Payloads served with Content-Type: application/x-chrome-extension to blend in with legitimate browser traffic

Impact Assessment

Host **10.1.17.215 (DESKTOP-L8C5GSJ)**, used by domain user **shutchenson (Steve Hutchenson)**, retrieved a PowerShell script and two additional payloads from external infrastructure, then began a sustained beaconing pattern that persisted for the observed duration of the capture. Kerberos ticket activity for the same account recurs throughout this window, indicating the account remained actively authenticated to the domain (**BLUEMOONTUESDAY**) while this activity was underway — this was not an idle or logged-off machine.

Two risks stand out:

- **Local execution risk:** a .ps1 file was explicitly downloaded. If executed, PowerShell provides broad capability for further payload staging, credential access, or lateral movement tooling — this capture alone does not confirm execution, only retrieval.
- **Persistent C2 risk:** the ~5-second beacon interval to 5.252.153.241 continued for the full reviewed capture window (over 40 minutes), and the secondary payload retrieval from a second host (199.232.214.172) suggests a multi-stage delivery chain rather than a single dropped file. This pattern is more consistent with an active, ongoing implant than a one-time, contained event.

This capture does not confirm whether the PowerShell script executed, what it did if run, or whether lateral movement occurred — those require endpoint (EDR/memory) forensics beyond what this pcap can show.

Recommendations

Immediate

- **Isolate the endpoint:** quarantine **10.1.17.215** via EDR network isolation or switch-port shutdown to halt further beaconing while preserving volatile memory for forensic capture.
- **Block the indicators:** block **5.252.153.241** and **199.232.214.172** at the perimeter firewall/proxy, along with the specific URI patterns identified (/api/file/get-file/*, /1517096937, /filestreamingservice/files/.*).
- **Contain the account:** disable **shutchenson** and force a credential reset once the endpoint is isolated, given the account was actively authenticated during the malicious activity.
- **Recover and analyze the script:** obtain a copy of 29842.ps1 (from the isolated host or via the blocked URL, in a sandboxed environment) to determine what the script does, and check for evidence it executed (PowerShell operational logs, AMSI logs, script block logging).

Long-Term

- Hunt enterprise-wide for connections to **5.252.153.241** and **199.232.214.172**, and for the specific URI patterns above.
- Alert on HTTP responses serving Content-Type: application/x-chrome-extension or outside expected extension-update flows.
- Enable PowerShell Script Block Logging and Constrained Language Mode where feasible, to reduce the impact of downloaded-script execution.
- Alert on repeated, fixed-interval HTTP requests returning 404 to the same host — a low-noise signature for polling-style C2 beacons.
- Review DNS/HTTP proxy logs for other hosts exhibiting the same /api/file/get-file/ or /filestreamingservice/ request patterns.

Conclusion

The investigation confirmed that Windows workstation **10.1.17.215 (DESKTOP-L8C5GSJ)**, MAC **00:d0:b7:26:4a:74**, used by domain account **shutchenson (Steve Hutchenson)**, retrieved a PowerShell script and two additional payloads from external hosts **5.252.153.241** and **199.232.214.172**, then engaged in sustained, fixed-interval HTTP beaconing for the remainder of the capture. Correlating LLMNR, Kerberos, and SAMR traffic with the HTTP download and beaconing activity provided sufficient evidence to positively identify the affected asset and its user, supporting a finding of compromise via a PowerShell-based downloader with ongoing C2 communication. The recommendations above should be actioned before the host is returned to service.

Appendix — Evidence Screenshots

Figure 1. HTTP-filtered view showing the initial malicious file request (GET /api/file/get-file/264872) from 10.1.17.215 to 5.252.153.241, followed in the same capture window by the .ps1 script download and repeated 404 beacon polling. The expanded Ethernet II / IPv4 headers on the selected frame confirm both the source IP (10.1.17.215) and MAC address (00:d0:b7:26:4a:74) of the infected host.

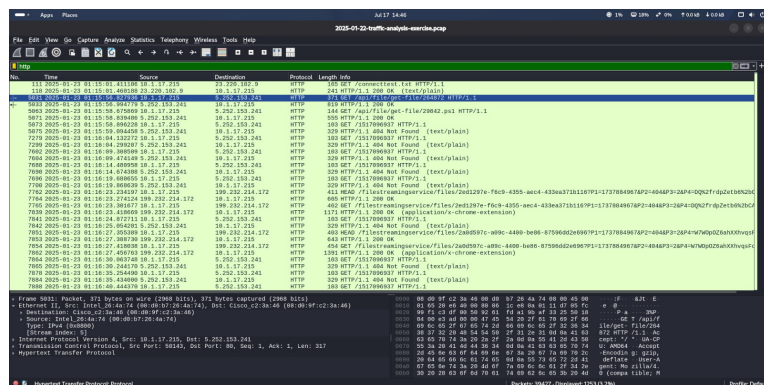


Figure 1

Figure 2. LLMNR query traffic (llmnr filter) from 10.1.17.215 resolving hostname DESKTOP-L8C5GSJ.

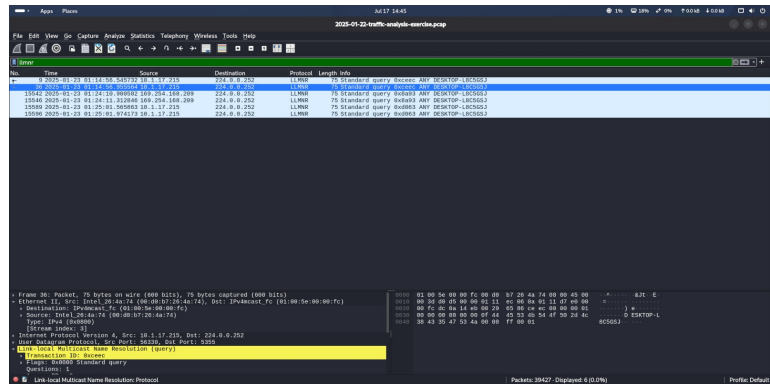


Figure 2

Figure 3. Kerberos traffic (kerberos.CNameString filter) identifying authenticated client principal shutchenson within the BLUemoontuesday realm.

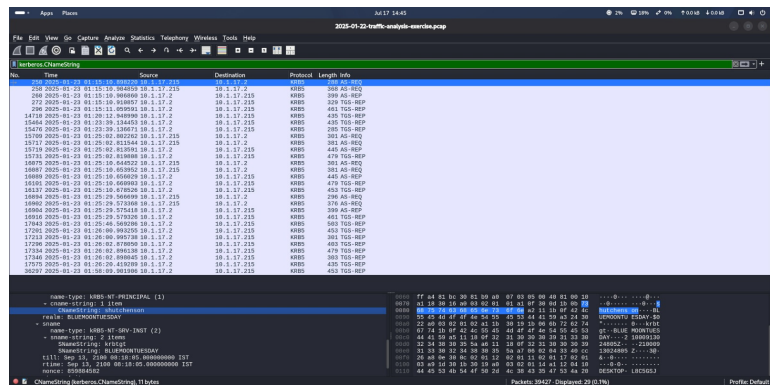


Figure 3

Figure 4. SAMR QueryUserInfo response (Info21) resolving account shutchenson to full name “Steve Hutchenson”.

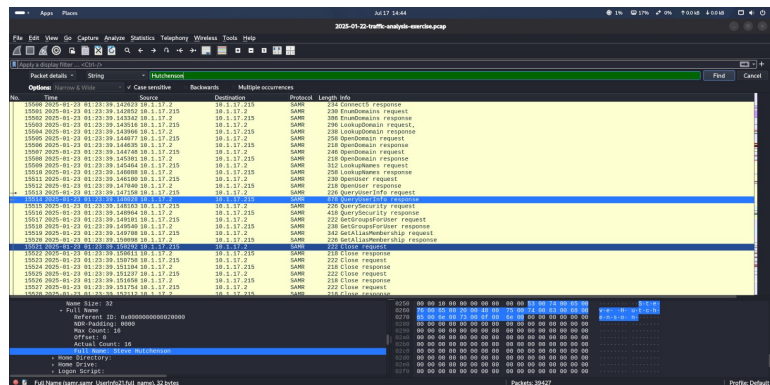


Figure 4

Report prepared by Lakshmeesh R as part of a SOC investigation portfolio exercise.